20261029 Midterm

for (MATH261) Discrete Mathematics

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Problem. Prove that $\sqrt{2}$ is irrational.

Solution. Suppose that $\sqrt{2} = p/q$ for relatively prime positive integers p, q. Then $p^2 = 2q^2$, so p^2 and hence p are even. Write $p = 2r$. Substitution gives $4r^2 = 2q^2$, or $q^2 = 2r^2$, so q is even as well. Thus 2 divides both p and q, contrary to $\gcd(p, q) = 1$.

□

Problem. Prove that there exist irrational real numbers x, y for which $x^y \in \mathbb{Q}$.

Solution. Consider $(\sqrt{2})^{\sqrt{2}}$. If this number is rational, choose $x = y = \sqrt{2}$. Otherwise choose $x = (\sqrt{2})^{\sqrt{2}}$ and $y = \sqrt{2}$; both are irrational, and

$$x^y = ((\sqrt{2})^{\sqrt{2}})^{\sqrt{2}} = (\sqrt{2})^2 = 2 \in \mathbb{Q}.$$

Thus either case supplies the required pair.

□

Problem. Prove that among any $n+1$ distinct positive integers not exceeding $2n$, where $n \in \mathbb{N}$, there are two such that one divides the other.

Solution. Each integer $m \in [2n]$ can be written uniquely as $m = 2^k u$, where $k \in \mathbb{N}_0$ and u is odd. There are exactly n possible odd parts $u \in [2n]$. By the Pigeonhole Principle, two of the selected integers have the same odd part, say $2^a u$ and $2^b u$. They are distinct, so we may assume $a < b$. Since

$$2^b u = 2^{b-a}(2^a u),$$

the first divides the second.

□

Problem. Prove that every sequence of $n^2 + 1$ distinct real numbers, where $n \in \mathbb{N}$, contains a strictly increasing or strictly decreasing subsequence of length $n + 1$.

Solution. Write the sequence as $a_1, \ldots, a_{n^2+1}$. For each i, let u_i and v_i be the lengths of the longest strictly increasing and decreasing subsequences ending at a_i, respectively. If neither kind has length $n + 1$, every pair (u_i, v_i) lies in $[n] \times [n]$, a set of size n^2.

But these $n^2 + 1$ pairs are distinct. For $i < j$, either $a_i < a_j$, in which case $u_j \geq u_i + 1$, or $a_i > a_j$, in which case $v_j \geq v_i + 1$: append a_j to a longest subsequence ending at a_i of the appropriate kind. This contradicts the Pigeonhole Principle.

□

Problem. Construct a bijection between $\mathbb{N}$ and $\mathbb{Z}$, and hence prove that $|\mathbb{N}| = |\mathbb{Z}|$.

Solution. Define $f\colon \mathbb{N} \to \mathbb{Z}$ by

$$f(n) = \begin{cases} 0, & n = 1, \\ k, & n = 2k \quad (k \in \mathbb{N}), \\ -k, & n = 2k+1 \quad (k \in \mathbb{N}). \end{cases}$$

Every integer occurs: $0 = f(1)$, $k = f(2k)$, and $-k = f(2k+1)$ for $k \in \mathbb{N}$. Hence f is surjective. The three cases have disjoint images, and within each case the rule is injective. Thus f is also injective and hence a bijection.

□

Problem. There are $2n$ people waiting in line for theater tickets costing 50 cents. Exactly n have a 50-cent coin and n have a one-dollar bill. The cashier initially has no change. How many orders allow every customer to buy a ticket without a shortage of change? Consider both indistinguishable and distinct people within each payment type.

Solution. Represent a customer paying 50 cents by $+1$ and one paying a dollar by -1. The cashier can give change at every step exactly when every initial partial sum is nonnegative.

There are n entries of each sign, so the admissible payment-type sequences are exactly the Catalan words. Their number is

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

This is the answer when people of the same payment type are indistinguishable. When all $2n$ people are distinct, the positions of each payment type can be filled in $n!$ ways, independently. The answer is therefore

$$(n!)^2 C_n = \frac{(2n)!}{n+1}.$$

□

Problem. Let $f_0 = 0$, $f_1 = 1$, and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$. Find an explicit formula for f_n.

Solution. The characteristic equation $r^2 - r - 1 = 0$ has distinct roots

$$\alpha = \frac{1+\sqrt{5}}{2}, \qquad \beta = \frac{1-\sqrt{5}}{2}.$$

Thus $f_n = A\alpha^n + B\beta^n$. The initial conditions give

$$A + B = 0, \qquad A\alpha + B\beta = 1.$$

Hence $B = -A$ and $A(\alpha - \beta) = 1$. Since $\alpha - \beta = \sqrt{5}$, we obtain

$$\boxed{f_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right).}$$

□

Problem. Solve

$$a_n = 6a_{n-1} - 9a_{n-2} \qquad (n \geq 2),$$

where $a_0 = 1$ and $a_1 = 6$.

Solution. The characteristic equation is $(r-3)^2 = 0$. Since 3 is a double root, the general solution is

$$a_n = (A + Bn)3^n.$$

The conditions $a_0 = 1$ and $a_1 = 6$ give $A = 1$ and $3(A + B) = 6$, whence $B = 1$. Therefore

$$\boxed{a_n = (n+1)3^n \qquad (n \geq 0).}$$

□

Problem. Solve

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3} \qquad (n \geq 3),$$

where $a_0 = 1$, $a_1 = -2$, and $a_2 = -1$.

Solution. The characteristic equation is

$$r^3 + 3r^2 + 3r + 1 = (r+1)^3 = 0.$$

Because -1 is a root of multiplicity three,

$$a_n = (A + Bn + Cn^2)(-1)^n.$$

Using the three initial conditions yields

$$A = 1, \qquad A + B + C = 2, \qquad A + 2B + 4C = -1.$$

Thus $B + C = 1$ and $B + 2C = -1$, so $B = 3$ and $C = -2$. Consequently,

$$\boxed{a_n = (1 + 3n - 2n^2)(-1)^n \qquad (n \geq 0).}$$

□

Problem. Find all solutions of $a_n = 3a_{n-1} + 2n$ for $n \geq 2$. Which solution satisfies $a_1 = 3$?

Solution. The homogeneous equation $a_n = 3a_{n-1}$ has solutions $A3^n$. Since the forcing term is linear, seek a particular solution $a_n^{(p)} = cn + d$. Substitution gives

$$cn + d = 3(c(n-1) + d) + 2n = (3c + 2)n + 3d - 3c.$$

Matching coefficients yields

$$c = 3c + 2, \qquad d = 3d - 3c,$$

so $c = -1$ and $d = -\frac{3}{2}$. Thus all solutions are

$$a_n = A3^n - n - \frac{3}{2}.$$

If $a_1 = 3$, then

$$3 = 3A - 1 - \frac{3}{2}, \qquad \text{so} \qquad A = \frac{11}{6}.$$

The required solution is

$$\boxed{a_n = \frac{11}{6}3^n - n - \frac{3}{2}.}$$

□

Problem. Find all solutions of

$$a_n = 5a_{n-1} - 6a_{n-2} + 7^n \qquad (n \geq 2).$$

Solution. The characteristic equation of the associated homogeneous recurrence is $(r-2)(r-3) = 0$, giving the homogeneous solution $A2^n + B3^n$. Since 7 is not a characteristic root, try a particular solution $a_n^{(p)} = C7^n$. Its substitution gives

$$C7^n = 5C7^{n-1} - 6C7^{n-2} + 7^n.$$

Dividing by 7^{n-2} and collecting terms,

$$49C = 35C - 6C + 49, \qquad 20C = 49, \qquad C = \frac{49}{20}.$$

Therefore all solutions are

$$\boxed{a_n = A2^n + B3^n + \frac{49}{20}7^n.}$$

□

Problem. Let $a_0 = 0$ and

$$a_n = \frac{2(n-1)}{n}a_{n-1} + \frac{1}{n} \qquad (n \geq 1).$$

Find an explicit formula for a_n by two different methods.

Solution 1. Multiplying by n suggests the substitution $b_n = na_n$. For $n \geq 1$ the recurrence becomes

$$b_n = 2b_{n-1} + 1, \qquad b_0 = 0.$$

Since $b_n + 1 = 2(b_{n-1} + 1)$, we have $b_n + 1 = 2^n(b_0 + 1) = 2^n$. Consequently,

$$\boxed{a_n = \frac{2^n - 1}{n} \quad (n \geq 1), \qquad a_0 = 0.}$$

□

Solution 2. The recurrence gives $a_1 = 1$. For $n \geq 2$, the coefficient $2(n-1)/n$ is nonzero, so the product–sum formula for first-order recurrences may be applied starting from a_1:

$$\begin{aligned} a_n &= a_1 \prod_{j=2}^{n} \frac{2(j-1)}{j} + \sum_{k=2}^{n} \frac{1}{k} \prod_{j=k+1}^{n} \frac{2(j-1)}{j} \\ &= \frac{2^{n-1}}{n} + \frac{1}{n}\sum_{k=2}^{n} 2^{n-k} \\ &= \frac{2^{n-1}}{n} + \frac{2^{n-1} - 1}{n} = \frac{2^n - 1}{n}. \end{aligned}$$

The products telescope; for example, $\displaystyle\prod_{j=k+1}^{n} (j-1)/j = k/n$. The first step $n = 1$ was handled separately because its recurrence coefficient is zero.

□

Problem. Let $m \in \mathbb{N}$. Prove the following identities for $|x| < 1$:

(a) $$(1+x)^{-m} = \sum_{n=0}^{\infty} \binom{m+n-1}{m-1} (-x)^n.$$

(b) $$(1-x)^{-m} = \sum_{n=0}^{\infty} \binom{m+n-1}{m-1} x^n.$$

Solution. For $n \geq 1$,

$$\begin{aligned}\binom{-m}{n} &= \frac{(-m)(-m-1)\cdots(-m-n+1)}{n!} \\ &= (-1)^n \frac{(m+n-1)!}{n!(m-1)!} \\ &= (-1)^n \binom{m+n-1}{m-1}.\end{aligned}$$

This identity also holds for $n = 0$, since both binomial coefficients are 1. The Generalized Binomial Theorem therefore gives

$$(1+x)^{-m} = \sum_{n=0}^{\infty} \binom{-m}{n} x^n = \sum_{n=0}^{\infty} \binom{m+n-1}{m-1} (-x)^n.$$

Replacing x by $-x$ yields (b). Both expansions converge for $|x| < 1$.

□

Problem. Use an ordinary generating function to solve

$$a_n = 5a_{n-1} - 6a_{n-2} \qquad (n \geq 2),$$

where $a_0 = 0$ and $a_1 = 2$.

Solution. Let $F(x) = \sum_{n \geq 0} a_n x^n$. Multiplying the recurrence by x^n and summing over $n \geq 2$ yields

$$F(x) - a_0 - a_1 x = 5x(F(x) - a_0) - 6x^2 F(x).$$

Substituting the initial conditions and factoring the denominator,

$$F(x) = \frac{2x}{1 - 5x + 6x^2} = \frac{2x}{(1-2x)(1-3x)}.$$

Since

$$2x = -2(1-3x) + 2(1-2x),$$

partial fractions give

$$\begin{aligned}F(x) &= -\frac{2}{1-2x} + \frac{2}{1-3x} \\ &= \sum_{n \geq 0} 2(3^n - 2^n) x^n.\end{aligned}$$

Comparing coefficients,

$$\boxed{a_n = 2(3^n - 2^n) \qquad (n \geq 0).}$$

□

Problem. Use an ordinary generating function to solve

$$a_n = -a_{n-1} + 6a_{n-2} \qquad (n \geq 2),$$

where $a_0 = 1$ and $a_1 = 3$.

Solution. Let $F(x) = \sum_{n\geq 0} a_n x^n$. Summation for $n \geq 2$ gives

$$F(x) - 1 - 3x = -x(F(x) - 1) + 6x^2 F(x),$$

and therefore

$$F(x) = \frac{1+4x}{1+x-6x^2} = \frac{1+4x}{(1-2x)(1+3x)}.$$

Because

$$1 + 4x = \frac{6}{5}(1+3x) - \frac{1}{5}(1-2x),$$

we obtain

$$\begin{aligned} F(x) &= \frac{6}{5(1-2x)} - \frac{1}{5(1+3x)} \\ &= \sum_{n\geq 0} \left(\frac{6}{5}\, 2^n - \frac{1}{5}(-3)^n \right) x^n. \end{aligned}$$

Comparing coefficients,

$$\boxed{a_n = \frac{6}{5}\, 2^n - \frac{1}{5}(-3)^n \qquad (n \geq 0).}$$

□

Problem. Use an ordinary generating function to solve

$$b_n = b_{n-1} + 2b_{n-2} + 2^n \qquad (n \geq 2),$$

where $b_0 = 2$ and $b_1 = 1$.

Solution. Let $F(x) = \sum_{n\geq 0} b_n x^n$. Because

$$\sum_{n\geq 2} 2^n x^n = \frac{1}{1-2x} - 1 - 2x,$$

the recurrence gives

$$F(x) - 2 - x = x(F(x) - 2) + 2x^2 F(x) + \frac{1}{1-2x} - 1 - 2x.$$

Thus

$$\begin{aligned} (1 - x - 2x^2)F(x) &= 1 - 3x + \frac{1}{1-2x}, \\ F(x) &= \frac{2 - 5x + 6x^2}{(1-2x)^2(1+x)}. \end{aligned}$$

Use the partial fraction decomposition

$$F(x) = \frac{A}{1-2x} + \frac{B}{(1-2x)^2} + \frac{C}{1+x}.$$

Equating numerators gives

$$2 - 5x + 6x^2 = A(1-2x)(1+x) + B(1+x) + C(1-2x)^2.$$

At $x = \frac{1}{2}$ this yields $1 = \frac{3}{2}B$, so $B = \frac{2}{3}$. At $x = -1$, it gives $13 = 9C$, so $C = \frac{13}{9}$. Finally $F(0) = 2$ implies

$$A = 2 - B - C = 2 - \frac{2}{3} - \frac{13}{9} = -\frac{1}{9}.$$

Consequently,

$$F(x) = -\frac{1}{9(1-2x)} + \frac{2}{3(1-2x)^2} + \frac{13}{9(1+x)}.$$

Finally, $[x^n](1-2x)^{-2} = (n+1)2^n$ gives

$$\boxed{b_n = -\frac{1}{9}\,2^n + \frac{2}{3}(n+1)2^n + \frac{13}{9}(-1)^n \qquad (n \geq 0).}$$

□

Problem. Let $(C_n)_{n\geq 0}$ be the Catalan numbers, with $C_0 = 1$ and

$$C_n = \sum_{j=0}^{n-1} C_j C_{n-1-j} \qquad (n \geq 1).$$

(a) Find the ordinary generating function $F(x)$ of the Catalan numbers.

(b) Using this generating function and the Generalized Binomial Theorem, prove

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

(c) Set $a_n = nC_n$ and find its ordinary generating function $G(x)$.

Solution.

(a) Let $F(x)$ be the ordinary generating function of the Catalan numbers. By the Cauchy product formula,

$$\begin{aligned} F(x) - 1 &= \sum_{n\geq 1} \left(\sum_{j=0}^{n-1} C_j C_{n-1-j} \right) x^n \\ &= x \sum_{m\geq 0} \left(\sum_{j=0}^{m} C_j C_{m-j} \right) x^m = xF(x)^2. \end{aligned}$$

Thus $xF(x)^2 - F(x) + 1 = 0$. Solving this quadratic equation,

$$F(x) = \frac{1 \pm \sqrt{1-4x}}{2x}.$$

The plus sign cannot give a formal power series with constant coefficient $F(0) = 1$. Therefore

$$\boxed{F(x) = \frac{1 - \sqrt{1-4x}}{2x}.}$$

(b) The Generalized Binomial Theorem gives

$$\sqrt{1-4x} = \sum_{j\geq 0} \binom{1/2}{j} (-4x)^j.$$

Hence

$$C_n = [x^n]F(x) = -\frac{1}{2}\binom{1/2}{n+1}(-4)^{n+1}.$$

For $n \geq 0$, expanding the generalized binomial coefficient gives

$$\binom{1/2}{n+1} = \frac{(1/2)(-1/2)\cdots(1/2-n)}{(n+1)!} = \frac{(-1)^n(2n-1)!!}{2^{n+1}(n+1)!}.$$

Thus, using $(2n)! = 2^n n!(2n-1)!!$ and $(-1)!! = 1$,

$$\begin{aligned} C_n &= \frac{2^n(2n-1)!!}{(n+1)!} = \frac{(2n)!}{n!(n+1)!} \\ &= \boxed{\frac{1}{n+1}\binom{2n}{n}}. \end{aligned}$$

(c) Termwise differentiation gives

$$G(x) = \sum_{n\geq 0} nC_n x^n = xF'(x).$$

Differentiate the identity $2xF(x) = 1 - \sqrt{1-4x}$:

$$2F(x) + 2xF'(x) = \frac{2}{\sqrt{1-4x}}.$$

Therefore

$$G(x) = xF'(x) = \boxed{\frac{1}{\sqrt{1-4x}} - \frac{1-\sqrt{1-4x}}{2x}}.$$

□

Problem. Let D_n be the number of derangements of an n-element set, with $D_0 = 1$, $D_1 = 0$, and

$$D_n = nD_{n-1} + (-1)^n \qquad (n \geq 2).$$

Find the exponential generating function of $(D_n)_{n\geq 0}$.

Solution. Let $F(x) = \sum_{n\geq 0} D_n \frac{x^n}{n!}$. Using the recurrence from $n = 2$, together with $D_0 = 1$ and $D_1 = 0$,

$$\begin{aligned} F(x) &= 1 + \sum_{n\geq 2} nD_{n-1}\frac{x^n}{n!} + \sum_{n\geq 2}(-1)^n\frac{x^n}{n!} \\ &= 1 + x(F(x)-1) + (e^{-x} - 1 + x) \\ &= xF(x) + e^{-x}. \end{aligned}$$

Therefore $(1-x)F(x) = e^{-x}$, and

$$\boxed{F(x) = \frac{e^{-x}}{1-x}}.$$

□

Problem. Let $B_0 = 1$ and let the Bell numbers satisfy

$$B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_k \qquad (n \geq 0).$$

Find the exponential generating function of $(B_n)_{n\geq 0}$.

Solution. Let $G(x) = \sum_{n\geq 0} B_n \frac{x^n}{n!}$. By termwise differentiation and the recurrence,

$$G'(x) = \sum_{n\geq 0} B_{n+1} \frac{x^n}{n!} = \sum_{n\geq 0} \left(\sum_{k=0}^{n} \binom{n}{k} B_k \right) \frac{x^n}{n!}.$$

The exponential generating function product rule gives

$$\begin{aligned} e^x G(x) &= \left(\sum_{j\geq 0} \frac{x^j}{j!} \right) \left(\sum_{k\geq 0} B_k \frac{x^k}{k!} \right) \\ &= \sum_{n\geq 0} \left(\sum_{k=0}^{n} \binom{n}{k} B_k \right) \frac{x^n}{n!}. \end{aligned}$$

Thus $G'(x) = e^x G(x)$ with $G(0) = 1$. Since a formal power series with constant coefficient 1 has a well-defined formal logarithm, we may integrate $(\log G(x))' = e^x$ term by term to obtain

$$\log G(x) = e^x - 1.$$

Hence

$$\boxed{G(x) = \exp(e^x - 1).}$$

□

Problem. For $n, k \in \mathbb{N}_0$, let $\left\{ {n \atop k} \right\}$ be the Stirling number of the second kind, where $\left\{ {0 \atop 0} \right\} = 1$ and $\left\{ {n \atop 0} \right\} = 0$ for $n \geq 1$.

(a) For fixed k, find the ordinary generating function $F_k(x)$ of $(\left\{ {n \atop k} \right\})_{n\geq 0}$.

(b) For fixed k, find the exponential generating function $G_k(x)$ of $(\left\{ {n \atop k} \right\})_{n\geq 0}$.

Solution.

(a) For $k \geq 1$, the recurrence $\left\{ {n \atop k} \right\} = k\left\{ {n-1 \atop k} \right\} + \left\{ {n-1 \atop k-1} \right\}$ for $n \geq 1$ implies

$$\begin{aligned} F_k(x) &= \sum_{n\geq 1} \left\{ {n \atop k} \right\} x^n \\ &= kxF_k(x) + xF_{k-1}(x). \end{aligned}$$

The first equality uses $\left\{ {0 \atop k} \right\} = 0$ for $k \geq 1$. Therefore

$$F_k(x) = \frac{x}{1-kx} F_{k-1}(x).$$

Since $F_0(x) = 1$, iteration gives

$$\boxed{F_k(x) = \frac{x^k}{\prod_{j=1}^{k}(1-jx)} \qquad (k \in \mathbb{N}_0).}$$

For $k = 0$, the product is empty and the formula gives $F_0(x) = 1$.

(b) For $k \geq 1$, the surjection-counting formula gives

$$k!\left\{ {n \atop k} \right\} = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} j^n \qquad (n \geq 1).$$

Since $\left\{ {0 \atop k} \right\} = 0$ for $k \geq 1$, multiply by $x^n/n!$ and sum over $n \geq 1$:

$$\begin{aligned} k!G_k(x) &= \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} \sum_{n \geq 1} \frac{(jx)^n}{n!} \\ &= \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} (e^{jx} - 1) \\ &= \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} e^{jx} = (e^x - 1)^k, \end{aligned}$$

where $\sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} = 0$ for $k \geq 1$. Consequently,

$$\boxed{G_k(x) = \frac{(e^x - 1)^k}{k!} \qquad (k \in \mathbb{N}_0).}$$

When $k = 0$, $\left\{ {n \atop 0} \right\} = 0$ for $n \geq 1$ and $\left\{ {0 \atop 0} \right\} = 1$, so $G_0(x) = 1$ agrees with this formula.

□